

Basavaraj Jakkannavar

Senior Data Scientist

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 Bengaluru, Karnataka, India

SKILLS

- **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Transformer Models, Generative AI, LLM Fine-tuning, Agentic AI Systems, Recommendation Systems
- **Programming:** Python, SQL, Java, PySpark
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, HuggingFace
- **Data Platforms:** Snowflake, BigQuery, Azure Synapse, AWS Athena, S3
- **MLOps & Infrastructure:** Azure ML, CI/CD, Docker, Kubernetes, Model Monitoring
- **Data Engineering:** ETL Pipelines, Airflow, Data Warehousing, Feature Engineering, PowerBI

EDUCATION

- **Birla Institute of Technology & Science (BITS)**

PhD Scholar (Data Science)
2025-2030

Master of Technology
Data Science
[Mar 2021 to Apr 2023]
- **Visvesvaraya Technological University (VTU)**
Bachelor of Engineering
Computer Science
[Aug 2011 to May 2014]
- **Karnataka Technical Education**
Technical Diploma
Computer Science
[Jun 2008 - Jul 2011]

PROFESSIONAL SUMMARY

Senior Data Scientist with 11 years of experience building large-scale machine learning systems and enterprise AI solutions across multiple industries. Proven expertise in deploying production ML systems, developing transformer-based NLP solutions, and designing scalable MLOps pipelines. Experienced in customer analytics, fraud detection, forecasting, recommendation systems, and generative AI applications. Recently focused on LLM-powered applications and agentic AI systems for enterprise automation.

WORK EXPERIENCE

Senior Data Scientist

ConnectWise LLPJan - 2024 - Present

- Architected a multi-agent AI system for automated ticket routing, where specialized agents predict attributes such as board, priority, contact, and company to route 2.1M+ support tickets per quarter.
- Developed a GEPA-based prompt optimization framework that iteratively evolves agent prompts using evaluation feedback, improving agent decision accuracy and robustness.
- Generated \$1.3M in incremental revenue by developing an LLM-powered partner intelligence platform that analyzed usage patterns, company attributes, and industry signals to identify upsell and cross-sell opportunities.
- Built a behavioral fraud detection system achieving >90% accuracy, processing millions of activity logs to identify suspicious patterns and proactively block scam accounts.

Senior Data Scientist

Brillio Technologies Pvt LtdMar 2021 - Jan - 2024

- Increased trial-to-paid conversion by 7% by designing customer propensity models leveraging behavioral and product usage features, enabling sales teams to prioritize high-probability leads at scale.
- Designed and deployed an end-to-end MLOps platform supporting 600+ production ML models.

Consultant - Machine Learning

Deloitte Consulting LLPJun 2017 - Mar 2021

- Built end-to-end machine learning solutions across domains including credit risk modeling, demand forecasting, marketing analytics, and NLP-based text classification.
- Developed predictive risk models for trade credit insurance using historical claims, sector indicators, and customer financial attributes to identify high-risk customers and industries and support underwriting decisions.
- Implemented time-series forecasting models to predict sales demand and support supply chain planning.

System Engineer - Data Analytics

Tata Consultancy ServicesMar 2015 - Jun 2017

- Developed and maintained ETL pipelines and data processing workflows using SQL and Java to support enterprise analytics and reporting systems.
- Designing data models for data warehouse, used JAVA/SQL to store, fetch and integrate data services to downstream applications.
- Worked as part of Master Data Management team to ensure Data Quality standards.

MAJOR PROJECTS

Agentic AI Ticket Routing Platform

Tech: LLMs, Multi-Agent Systems, Prompt Optimization (GEPA), Snowflake, Confidence Learning for Noise Detection, Pytorch.

- Designed and deployed an agentic AI platform for automated ticket routing, where multiple specialized agents predict key ticket attributes (board, agreement, company, contact, urgency) to process 2.1M+ support tickets per quarter.
- Developed a GEPA-based prompt evolution framework leveraging GPT-5-mini as the learner and GPT-5 as the reflective model to continuously improve prompt quality and agent decision accuracy.
- Implemented a multi-agent architecture enabling collaborative reasoning across agents to determine the optimal routing configuration.
- Deployed the system to 60+ Managed Service Providers (MSPs) and currently scaling the platform for 500+ additional enterprise customers

Scammer Detection System for Trial Account Abuse

Tech: ML Classification, Fraud Detection, Outlier Detection, LLMs Reasoning, PowerBi reporting

- Built a fraud detection system to identify suspicious trial accounts created to exploit a remote access software platform.
- Developed behavioral models analyzing activity logs, trial usage patterns, and time-to-purchase signals to detect accounts likely associated with scam activity.
- Implemented rule-based anomaly detection to flag accounts originating from suspicious IP ranges, geographies, and abnormal behavioral patterns.
- Combined machine learning and LLM-based reasoning with rule-based detection to automatically identify and restrict fraudulent accounts.

Strategic Enrolment & Student Price Sensitivity Analysis

Tech: Classification, MLOps at Scale, Model Ensembling, Pipelining, Model Exapliability, PowerBI

- Designed machine learning models to predict student enrollment probability using socio-economic, geo-demographic, and inquiry data, enabling universities to prioritize high-potential applicants.
- Developed optimization models for financial aid allocation, helping institutions determine the optimal scholarship strategy to increase enrollment rates and maximize institutional revenue.
- Scaled the platform to support 300+ universities, with 600+ models deployed in production for batch inference and enrollment decision support.

Transformer-Based Real Estate Recommendation Engine

Tech: PyTorch, Sequence Modeling, Recommendation Systems, Transformers.

- Developed a transformer-based recommendation engine to generate context-aware property recommendations for a real estate listings platform.
- Built a sequence-to-sequence model that takes a sequence of user interaction listing IDs and predicts the next likely listings, enabling personalized recommendations.
- Leveraged attention mechanisms to capture relationships across listing attributes and user interaction history, improving recommendation relevance.

Trade Insurance Risk Prediction and Portfolio Risk Scoring

Tech: Risk Modelling, Classification, Python, Sagemaker

- Built predictive models for trade insurance underwriting to estimate claim occurrence, claim frequency, and expected loss ratios for insurance applicants.
 - Leveraged historical claims data, applicant characteristics, and industry-level signals to model risk exposure and forecast potential insurance losses.
 - Designed a risk scoring framework to categorize applicants into risk segments, supporting improved underwriting decisions and portfolio risk management.
 - Generated industry risk profiles highlighting high-risk sectors (e.g., electronics) compared to lower-risk segments (e.g., consulting services).
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